zygote into a multicellular structure called blastocyst.

- **Implantation:** Attachment of blastocyst to the uterine wall.
- **Placentation:** Formation of placenta which is the intimate connection between foetus and uterine wall of the mother for exchange of nutrients.
- **Gastrulation:** Process by which blastocyst is changed into a gastrula with three primary germ layers.
- **Organogenesis:** Formation of specific tissues, organs and organ systems from three germ layers.
- **Parturition:** Expulsion of the foetus from the mother's womb.

These functions are carried out by the primary and accessory reproductive organs. The primary reproductive organs namely the ovary and testis are responsible for producing the ova and sperms respectively. Hormones secreted by the pituitary gland and the gonads help in the development of the secondary sexual characteristics, maturation of the reproductive system and regulation of normal functioning of the reproductive system. The accessory organs help in transport and to sustain the gametes and to nurture the developing offspring.

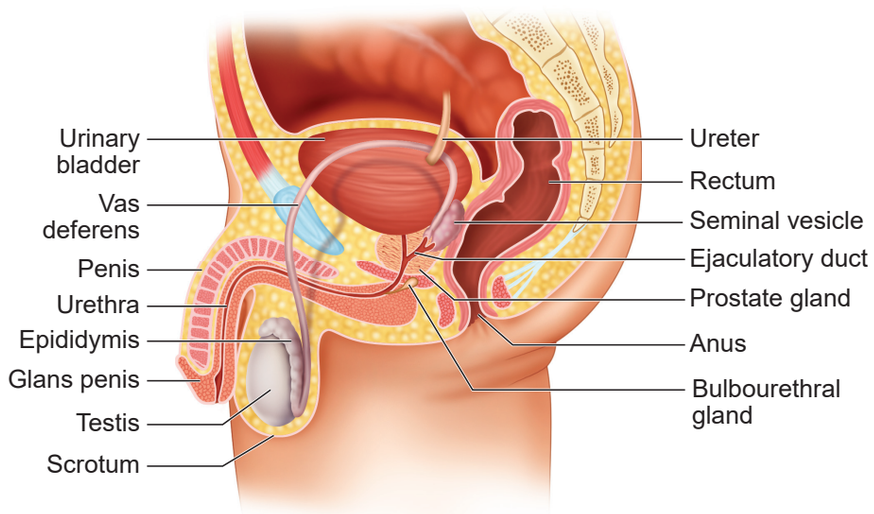


Fig. 2.1 Male reproductive system

2.1 Human reproductive system

The male reproductive system comprises of a pair of testes, accessory ducts, glands and external genitalia (**Fig. 2.1**).

Testes are the primary male sex organs. They are a pair of ovoid bodies lying in the scrotum (**Fig.2.2 a**). The scrotum is a sac of skin that hangs outside the abdominal cavity. Since viable sperms cannot be produced at normal body temperature, the scrotum is placed outside the abdominal cavity to provide a temperature 2-3°C lower than the normal internal body temperature. Thus, the scrotum acts as a **thermoregulator** for spermatogenesis.

Each testis is covered by an outermost fibrous **tunica albuginea** and is divided by septa into about **200 - 250 lobules** each containing 2-4 highly coiled testicular tubules or seminiferous tubules. These highly convoluted tubules which form 80 percent of the testicular substance are the sites for sperm production.

The stratified epithelium of the seminiferous tubule is made of two types of cells namely sertoli cells or nurse cells and spermatogonic cells or male germ cells. **Sertoli cells** are elongated and pyramidal and provide nourishment to the sperms till maturation. They also secrete **inhibin**, a hormone which is involved in the negative feedback control of sperm production. Spermatogonic cells divide meiotically and differentiate to produce spermatozoa.

Interstitial cells or Leydig cells are embedded in the soft connective tissue surrounding the seminiferous tubules. These cells are endocrine in nature and are

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